

Continuity (1.6)

Today's Target



A1. Rates of Change

"I can determine algebraically if a function is continuous, or not. I can identify, with an algebraic explanation, all discontinuities. I can apply what I have learned in familiar and unfamiliar settings."

Recall:

continuity.

(simply put)

continuity.

(simply put)

Non-stop $\rightarrow D: \{x \in \mathbb{R}\}$

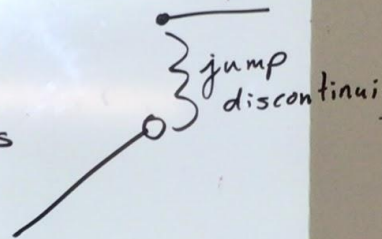
- No jump discontinuities
- No vert. asymptotes
- No holes

"A iff B"

$\hookrightarrow A$ "if and only if" B

$$A \iff B$$

$\downarrow$
iff



"A iff B"

continuity

(formally stated)

A function f is continuous at $x = a$ iff

1. $f(a)$ is defined

2. $\lim_{x \rightarrow a} f(x)$ exists

3. $\lim_{x \rightarrow a} f(x) = f(a)$

Example

Algebraically, test the continuity at $x = 0$. If it is discontinuous, explain why.

a) $f(x) = 3x - 2$

Handwritten work on a piece of paper showing the algebraic test for continuity of the function $f(x) = 3x - 2$ at $x = 0$.

At the top, the function is defined: $f(x) = 3x - 2$.

Step 1: Evaluate the function at $x = 0$.
$$\textcircled{1} \quad f(0) = 3(0) - 2$$
$$= -2$$

Step 2: Evaluate the limit of the function as x approaches 0 from the left.
$$\textcircled{2} \quad \lim_{x \rightarrow 0^-} 3x - 2$$
$$= -2$$

Step 3: Evaluate the limit of the function as x approaches 0 from the right.
$$\lim_{x \rightarrow 0^+} 3x - 2$$
$$= -2$$

Step 4: Conclude the continuity test.
$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$
$$\therefore f(x) \text{ is continuous}$$
$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

Example

Algebraically, test the continuity at $x = 0$. If it is discontinuous, explain why.

$$\text{b) } H(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

a) $H(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$
 $H(0) = 0$

b) $\lim_{x \rightarrow 0^+} H(x) = 1$
 $\lim_{x \rightarrow 0^-} H(x) = 0$

$\therefore \lim_{x \rightarrow 0^+} H(x) \neq \lim_{x \rightarrow 0^-} H(x)$
 $\therefore$ the limit does not exist
 $\therefore H$ is not continuous

Discussion: How would you algebraically discuss the continuity of the square root function at its “starting point”?

Example: Find any and all discontinuities for the following function, do not refer to the graph.

$$h(x) = \frac{x^2 - 16}{x^2 + 9x + 20}$$

This solution does not
reference the rules 2+

$$\begin{aligned} h(x) &= \frac{x^2 - 16}{x^2 + 9x + 20} \\ &= \frac{(x+4)(x-4)}{(x+5)(x+4)} \rightarrow \text{hole at } x = -4 \\ h(x) &= \frac{(x-4)}{(x+5)} \quad x \neq -5 \rightarrow \text{VA} \\ \therefore \text{Discontinuity is at } x = -5 \text{ \& } x = -4. \end{aligned}$$

$$h(x) = \frac{x^2 - 16}{x^2 + 9x + 20}$$

$$= \frac{(x-4)(\cancel{x+4})}{(x+5)(\cancel{x+4})} \quad \begin{array}{l} x \neq -4 \\ x \neq -5 \end{array}$$

$$= \frac{x-4}{x+5}$$

$\therefore f(x)$ is undefined at $x = -4$
and $x = -5$

$\therefore f(x)$ is discontinuous at $x = -4$
and $x = -5$

Put forth a "continuous" effort, which means no "breaks"!

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[Solutions to 15, 16, 17](#)